Bharat Singh Bhati

Machine Vision and IoT

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SUMMARY

Machine vision and IoT developer with a B.Tech in Computer Science (2026) and four years at RAIOT Labs building prototypes on Raspberry Pi and ESP32. Works in Python, C++ and OpenCV, making things that run offline on small hardware, like a Hinglish voice assistant and a mood light with gas alerts. Two-time IEEE hackathon winner.

EXPERIENCE

IoT Developer · RAIOT Labs, Amity University Jaipur 2022 – 2026

- Built 5 IoT prototypes in Python on Raspberry Pi, including a robotic arm and an air-quality monitor.
- Mentored 12+ juniors in Python, embedded systems and working with sensors.
- Presented 4 working prototypes at university tech exhibitions.

PROJECTS

Youru · Offline Hinglish Voice Assistant Python, Faster-Whisper, llama.cpp

- Offline voice assistant that wakes on its name, understands Hinglish, uses a local LLM and talks back.
- Trained my own wake-word model (95%+ accuracy in a quiet room) and a voice lock so it only obeys me.
- Got the LLM from 6 to about 89 tokens/s on a 4 GB laptop GPU; prompt caching saves ~1.3 s on each reply.
- Hooked up 30 voice-controlled desktop actions and brought screen OCR down from 6.2 s to 2.1 s.

LumiSense · Smart Lighting and Air-Quality Monitor Arduino, C++, MQTT

- Smart room lighting that changes colour with your mood, pulses to music and sets a calm ambient glow.
- Built-in gas sensing: if CO or CO_2 gets too high, the lights flash red and a buzzer goes off.

3D Portfolio Website · three.js, Python **bharatsingh7.github.io**

- A 3D room to explore, with a robot that answers questions about my work and a drone that waters plants.
- Reads my resume PDF and rebuilds itself on every push, and scales down its graphics on slower phones.

MacPhantom · Wi-Fi Network Scanner and MAC Spoofer Python, Netdiscover, Sockets

- Scans the Wi-Fi network I am on, finds every connected device and lists their MAC addresses.
- Changes my own MAC or takes on another device's, to test how easily MAC filtering can be bypassed.

SKILLS

Languages:	Python, C++, Java, SQL
Machine Vision and AI:	OpenCV, YOLO, Faster-Whisper, Ollama, NumPy, Pandas
Embedded and IoT:	ESP32, Arduino, Raspberry Pi, MQTT, Sensor Fusion, Edge Computing
Systems:	Linux (Arch, Debian), Docker, Git, Bash
Concepts:	REST APIs, Flask, OOP, DSA, DBMS

EDUCATION

B.Tech. Computer Science and Engineering · Amity University Rajasthan, Jaipur CGPA: 7.9/10 · 2022 – 2026

ACHIEVEMENTS

1st Place, IEEE Hackathon 2024 · **2nd Place**, IEEE Hackathon 2023
Lead Organizer, Sun Bulls Future Energies Pvt. Ltd. event · Participant, IIT Roorkee Cognizance 2025

CERTIFICATIONS

Prompt Engineering for Developers, DeepLearning.AI (Andrew Ng)
Robotron Robotics Course, Tech Analogy (DPIIT Recognised)